Guanghan Wang

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Research Interests

Diffusion language models and large language models more broadly: pretraining efficiency and scaling laws, reinforcement learning for reasoning, efficient inference and inference-time scaling.

Education

Cornell University (Cornell Tech), New York, NY — Aug 2024 – Present
Ph.D. Candidate in Computer Science (expected 2028) Advisor: Volodymyr Kuleshov

Tsinghua University, Beijing, China — Sep 2020 – Jun 2024
B.Eng. in Computer Science and Technology GPA: 3.93/4.00

Research and Industry Experience

Applied Scientist Intern (extended to 12 months), *Amazon*, Palo Alto, CA — Feb 2026 – Feb 2027

- Led a systematic study of diffusion vs. autoregressive language model pretraining efficiency, evaluating approximately **1,700 training runs** across ten model variants under matched compute budgets.
- Extended the training framework to support **block diffusion** and **encoder–decoder diffusion (E2D2)** architectures; fit scaling laws using exact likelihood across architecture choices and block sizes.
- Showed that the previously reported **16**× pretraining compute gap narrows to **8**× at block size 32 and **3**× at block size 4 under the revised protocol (arXiv 2026).

Graduate Research Assistant, *Cornell Tech*, Advisor: Volodymyr Kuleshov — Aug 2024 – Present

- **RL post-training for diffusion LMs** (d2, ICML 2026). Developed a GRPO-style post-training framework for diffusion LMs based on trajectory likelihood estimation, with two estimators: **d2-AnyOrder** (exact, single-pass) and **d2-StepMerge** (tunable compute–accuracy trade-off). On LLaDA-8B-Instruct, raised Sudoku accuracy from 11.8% to **91.9%**, Countdown from 19.9% to **56.6%**, GSM8K from 75.7% to **85.0%**, and MATH500 from 35.4% to **41.6%**, outperforming d1 and wd1 and setting a new state of the art for diffusion LMs.
- **Efficient inference and inference-time scaling for discrete diffusion** (ReMDM, NeurIPS 2025). Developed a remasking sampler for masked diffusion models that lets already-decoded tokens be revised, which both opens a **new axis of inference-time scaling**—the number of sampling steps under a fixed sequence length—and yields **better samples at fewer steps**, all without retraining; unifies existing predictor–corrector samplers as special cases. Achieved state-of-the-art generation quality among discrete diffusion models on text, molecules, and images.
- **Guidance and uniform-state diffusion** (ICLR 2025). Implemented and benchmarked classifier-free and classifier-based guidance for masked and uniform discrete diffusion across text and molecular tasks.

Visiting Research Student, *Carnegie Mellon University*, Jian Ma Lab — Jun 2023 – Dec 2023

- Built interpretable multiscale Transformers to analyze cell–cell interactions from spatial transcriptomics data (STEAMBOAT, RECOMB 2025).

Publications and Preprints (*: equal contribution; †: equal advising)

Selected

- **Revisiting the 16× Pretraining Compute Gap Between Diffusion and Autoregressive Language Models** [code]. **Guanghan Wang**, Bing He, Zhan Shi, Benoit Dumoulin, Hanqing Lu†, Volodymyr Kuleshov†. *arXiv preprint, 2026*
- **d2: Improving Reasoning in Diffusion Language Models via Trajectory Likelihood Estimation** [code]. **Guanghan Wang**, Gilad Turok, Yair Schiff, Marianne Arriola, Volodymyr Kuleshov. *ICML 2026*
- **Remasking Discrete Diffusion Models with Inference-Time Scaling** [code]. **Guanghan Wang***, Yair Schiff*, Subham Sekhar Sahoo, Volodymyr Kuleshov. *NeurIPS 2025*
- **Simple Guidance Mechanisms for Discrete Diffusion Models** [code]. Yair Schiff*, Subham Sekhar Sahoo*, Hao Phung*, **Guanghan Wang***, Sam Boshar, Hugo Dalla-Torre, Bernardo P. de Almeida, Alexander Rush, Thomas Pierrot, Volodymyr Kuleshov. *ICLR 2025*

Additional

- **Clock Diffusion: Efficient Semi-Autoregressive Continuous Diffusion Language Models**.
  Yair Schiff, Omer Belhasin, Roy Uziel, Matan Rusanovsky, Ran Zilberstein, Marianne Arriola, Gilad Turok, **Guanghan Wang**, Volodymyr Kuleshov†, Michael Elad†. *arXiv preprint, 2026*
- **Learn from Your Mistakes: Self-Correcting Masked Diffusion Models**.
  Yair Schiff*, Omer Belhasin*, Roy Uziel, **Guanghan Wang**, Marianne Arriola, Gilad Turok, Ran Zilberstein, Michael Elad†, Volodymyr Kuleshov†. *NeurIPS 2026*
- **The Diffusion Duality**.
  Subham Sekhar Sahoo, Justin Deschenaux, Aaron Gokaslan, **Guanghan Wang**, Justin Chiu, Volodymyr Kuleshov. *ICML 2025*
- **STEAMBOAT: Attention-Based Multiscale Delineation of Cellular Interactions in Tissues**.
  Shaoheng Liang, Junjie Tang, **Guanghan Wang**, Jian Ma. *RECOMB 2025*
- **Language-Assisted 3D Feature Learning for Semantic Scene Understanding**.
  Junbo Zhang, Guofan Fan, **Guanghan Wang**, Zhengyuan Su, Kaisheng Ma, Li Yi. *AAAI 2023*

Technical Skills

Languages & frameworks: Python, C/C++, PyTorch, Hugging Face Transformers/TRL, verl, PyTorch Lightning.
Large-scale training: multi-node GPU training with FSDP and DeepSpeed, scaling-law experiment pipelines ($\sim$1,700 runs), mixed-precision training.
Inference: custom diffusion samplers, vLLM.
Research expertise: LLM pretraining and scaling laws, RL post-training, diffusion models, inference-time scaling.

Selected Honors and Awards

- **Gold Reviewer Award**, ICML 2026.
- **Outstanding Graduate**, Tsinghua University, 2024.
- **Scholarship of Academic Excellence** (top 1% of department), Tsinghua University, 2021–2023.

Academic Service

- **Reviewer:** ICLR 2026–2027; ICML 2026; NeurIPS 2026; AAAI 2026; TMLR.
- **Invited talk:** Remasking Discrete Diffusion Models with Inference-Time Scaling, ASAP Seminar, 2025.
- **Teaching Assistant:** CS 5785 Applied Machine Learning, Cornell, Fall 2024.